AquaNode R0.3-EC | Freshwater EC estimate

Experimental assembly / no SHT40 / no new carrier copper / calibration mandatory / 29 October 2026

REMOVE R3 (4.7k) FROM CARRIER. J7 IS NOW EC, NOT DS18B20.

1. SEN0244 analog channel

J7.2 / 3V3 -> M1 VCC (+) | J7.1 / GND -> M1 GND (-)

M1 OUT (A) -> R21 1k -> J7.3 / EC_FILTERED -> existing R4 100R -> GPIO1 / D0

C19 100nF: EC_FILTERED to GND. Existing D2 ESD protection remains on EC_FILTERED.

R3 footprint remains EMPTY. Without removal, the old pull-up corrupts the analog reading.

ADC1, 11 dB attenuation; use calibrated mV. Sensor powered continuously from 3V3.

2. Move the existing DS18B20 probe to GPIO43 / D6

TP3 / 3V3 -> M2 VDD. TP6 / GND -> M2 GND.

J1.7 / D6 / GPIO43 -> R20 100R -> M2 DQ / DS_CABLE.

R19 4.7k: 3V3 to J1.7 (DS_LOCAL). Reuse the removed R3 if practical.

D3 PESD3V3S1BA: DS_CABLE to GND. All new parts on insulated internal perfboard.

Use USB CDC logging. Do not enable UART0 TX on GPIO43. No parasitic probe power.

Acceptance conditions

Measure sensor supply under load: must remain within 3.3-5.5 V. Check input levels before connecting GPIO.

EC output stays null until calibration; use only the independently validated freshwater interval.

pH/EC electrical interference is not resolved by this low-cost circuit. Validate separately before joint immersion.

Page 2 is the ORIGINAL R0.3 carrier reference. Apply the above changes; native CAD contains the new assembly.

Native ERC/DRC, ESP32 compilation and powered tests are pending. See EC_integration_ru.md.

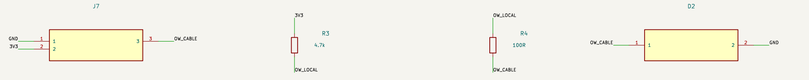
1. MCU & Radio Interface



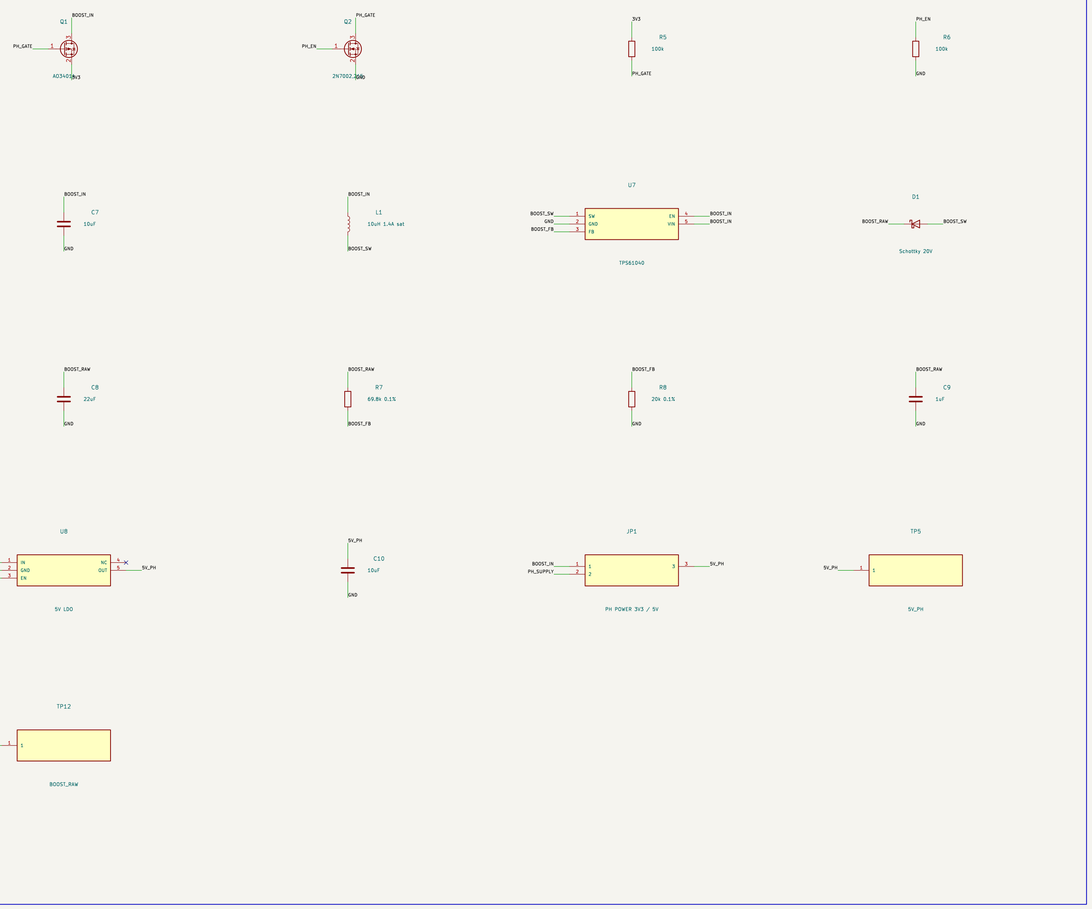
2. Battery & Switch



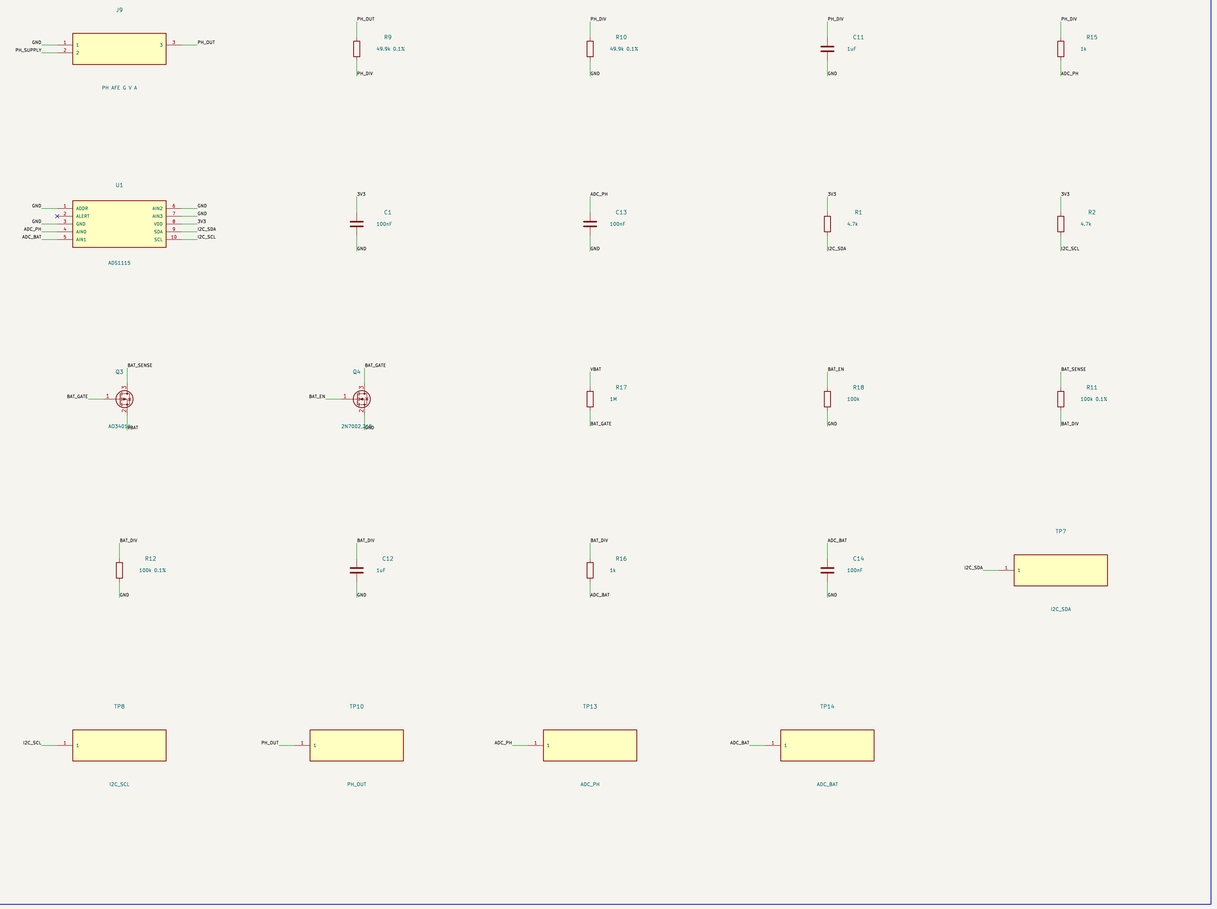
5. 1-Wire Temperature



3. Switched 5V Boost + LDO



4. Analog Front-End & ADC



R0.3: J9 = GND / VCC / OUT. Assembly A: R9=R10=49.9k. Assembly B (0..3V AFE): R9=10k, R10=DNP.
Wio connects to XIAO through B2B ONLY. Never bridge Wio left row to XIAO D0-D5.
16 SPS + RC reduces ripple, not DC input-loading error; see Analog_calculations_ru.md.
PCB is a reviewed prototype design. Mechanical fit and powered tests remain required.

Sheet: /	
File: AquaNode.kicad_sch	
Title: AquaNode R0.3 — PROTOTYPE DESIGN	
Size: A1	Date: 2026-10-20
KiCad E.D.A. 8.0.9	**Rev: R0.3**
	Id: 1/1